PETROCHEMICAL ENGINEERING S.A.
INSTRUMENTATION & CONTROL DEPT.

ISA-TR20.00.01 / ISA-20

DATA SHEET - ON-OFF VALVE

BALL VALVE

TAG NO.
HS-1200

REV: 1 HOJA: 1 / 1

PROYECTO	SERVICIO / FUNCIÓN	FECHA	ESTADO
REFINERY EXPANSION PROJECT UNIT 400	ON-OFF EMERGENCY SHUTDOWN (ESD)	30/04/2026	APPROVED FOR CONSTRUCTIO

P&ID REF.	LÍNEA / UBICACIÓN	ÁREA / PLANTA	O. COMPRA
P&ID-402-12-001	4"-HC-4021-A1A-H	COMPLEX SECTOR B - HYDRO	PO-88231-VALV

GENERAL

100	TAG NUMBER		HS-1200
101	Service Description		Light Naphtha Feed Cutoff
102	Revision / Status		Rev 1 / Approved
103	P&ID Reference		P&ID-402-12-001
104	Line Number		4"-HC-4021-A1A-H
105	Ambient Temperature Range	°C	-10 / +45
106	Hazardous Area Classification		Class I, Zone 1, Gr IIB+H2 T4
107	SIL Requirement		SIL 2 Certified (IEC 61508)

CONDITIONS

#	CONDICIÓN DE PROCESO	UNIDAD	MÍN	NORM	MÁX
201	Flow Rate	kg/h	1200	**3062**	4500
202	Inlet Pressure (P1)	bar g	8.5	**10.0**	12.4
203	Operating Temperature	°C	15	**42**	65
204	Design Pressure	bar g	-	**19.6**	-
205	Design Temperature	°C	-	**120**	-
206	Operating Density	kg/m³	690	**715**	730
207	Dynamic Viscosity	cP	0.45	**0.52**	0.61
208	Fluid State & Name			Liquid / Treated Naphtha	
209	Corrosive Components			H2S traces < 50 ppm, NACE MR0175	

LINE SPEC

301	Pipe Nominal Size	inch	4" (DN100)
302	Pipe Schedule & Wall		SCH 40 / 6.02 mm
303	Pipe Material		ASTM A106 Gr. B
304	End Connection Type		Flanged ASME B16.5
305	Flange Rating & Facing		Class 300 / Raised Face (RF)
306	Flange Finish	µin	Ra 125 - 250 AARH

VALVE BODY

401	Manufacturer & Model		M/S (1) / Trunnion Mounted
402	Body Style / Construction		2-Piece Split Body, Full Bore
403	Body Material		ASTM A216 Gr. WCB (Carbon Steel)
404	Ball & Stem Material		ASTM A182 F316 / 17-4PH
405	Seat Material & Insert		Devlon V-API / Primary PEEK
406	Stem Packing / Seals		Graphite Fire-Safe + Viton AED O-rin
407	Fire Safe Certified		Yes, API 607 7th Edition / ISO 10497
408	Shutoff Leakage Class		API 598 Rate A / ISO 5208 Rate A (0 l
409	Face to Face Standard		ASME B16.10 Long Pattern
410	NACE MR0175 / ISO 15156		Compliant

ACTUATOR

501	Manufacturer / Model		M/S (1) / Pneumatic Heavy Duty
502	Actuator Type		Scotch Yoke / Rack & Pinion
503	Actuator Action		Spring Return (Single Acting)
504	Fail Position on Loss of Air		FAIL CLOSE (FC)
505	Air Supply Min / Max	bar g	4.5 min / 7.0 max
506	Design Sizing Safety Factor		1.50 at min air supply
507	Max Stroking Time (Close)	sec	< 2.5 seconds (ESD)
508	Manual Override		Declutchable Manual Gear Operator
509	Body / Cylinder Material		Hard Anodized Aluminum / Epoxy co

SOLENOID

601	Manufacturer / Model		ASCO 551 Series (or equal)
602	Configuration & Action		3/2 Way, Direct Acting, De-energize t
603	Power Supply		24 VDC (Nominal) 3.5 W)
604	Explosion Proof Rating		Ex db IIC T6 Gb / IP66
605	Body & Internal Material		SS316 Stainless Steel
606	Electrical Connection		1/2" NPTF Conduit
607	Manual Reset / Test		Local push-button manual test lever

SWITCHES

701	Switchbox Manufacturer / Model		Westlock / TopWorx D-Series
702	Sensor Element Type		Inductive Proximity Sensors (NAMUR
703	Switches Quantity		2 (1 Open + 1 Closed Indication)
704	Visual Position Indicator		Beacon High-Visibility 3D Yellow/Bla
705	Enclosure Protection		NEMA 4X / IP67, Ex ia IIC T6
706	Terminal Strip Points		8-Point screw terminals

ACCESSORIES

801	Air Filter Regulator (AFR)		Included, SS316 with 5µm filter & ga
802	Quick Exhaust Valve		Yes, High-flow 1/2" SS316 body
803	Tubing & Fittings Material		ASTM A269 TP316L, Swagelok / Park
804	Breather / Exhaust Bug Screen		Installed on all exhaust ports (SS316)
805	Mounting Hardware & Bracket		ISO 5211 Standard, 304/316 Stainless

NOTAS

(900) (1) M/S: Manufacturer Standard compliant with project specifications SPEC-INST-400-01.
(901) (2) Valve assembly actuator and solenoid must be fully integrated and tested in factory per API 598.
(902) (3) Fire-safe certificate according to API 607 / ISO 10497 must be submitted with documentation

PREPARED BY	CHECKED BY	APPROVED BY	CLIENT APPROVAL
28/04/2026	29/04/2026	30/04/2026	02/05/2026
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